

Task: Image resizing.

Candidates will be evaluated for their

- Understanding of OpenCV
- Writing optimal c/c++ code.
- Debugging and performance optimization.

Step 1: Image resizing using [OpenCV](#) “cv::resize” function. Generate 3 resized outputs using 3 different interpolation methods for a given input image.

void [cv::resize](#) ([InputArray](#) src, [OutputArray](#) dst, [Size](#) dsize, double fx=0, double fy=0, int interpolation=[INTER_LINEAR](#))

Use: “INTER_NEAREST”, “INTER_LINEAR” and “INTER_CUBIC” as interpolation flags to generate 3 different outputs.

Input image:

<https://avproglobal.egnyte.com/fl/V6JZASnoul>

Download G178_2-1080.BMP from above link.

Step 2:

- Measure timing for 1000 iterations for each of these 3 different methods.

Time taken for 1000 iteration using INTER_NEAREST : [] ms

Time taken for 1000 iteration using INTER_LINEAR : [] ms

Time taken for 1000 iteration using INTER_CUBIC : [] ms

Step3:

- Write your own c/c++ function equivalent to cv::resize.
- Replace original cv::resize with your own function.
- Compare output for its correctness.
- Measure performance (timing) for all 3 methods of interpolation.

Notes:

- In step 2 and 3 you have to report the timing of only the resize function.
- You have to use a complete cpu to maximize performance. Don't run any compute intensive workload while measuring performance.
- For all resize, new_height = original_height / 2 and new_width = original width / 2.

Submission Instructions:

- Provide a clear README.md explaining how to build and run the application locally.
- Include any necessary dependencies and installation instructions.
- Include comments where necessary.
- Submit the assignment via a Git repository or a compressed file.